

The Topology of Reachable Nature: Persistent Homology and the Resilience of Green-Space Access in Cities of Contrasting Morphology

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Abstract

Sustainable Development Goal 11.7 calls for universal access to green and public space, yet accessibility is almost always quantified as a static snapshot that says nothing about how access degrades when the street network is disrupted. We introduce a *topological resilience metric* for accessibility: the Wasserstein distance between the persistent-homology diagram of a city’s green-space accessibility field before and after network disruption, traced as a function of disruption intensity ρ . The area under this degradation curve and its critical transition ρ^* summarise how robustly a place keeps its population close to nature. Applying the metric to five morphologically contrasting cities (İstanbul, Barcelona, Amsterdam, Bogotá, Phoenix) at the level of 2,603 intra-urban districts, we find that street-network morphology is a strong, statistically significant predictor of green-space access resilience: in a district-level regression with city fixed effects, street circuitry, orientation entropy, mean segment length and intersection density are all significant ($p < 0.01$; three at $p < 10^{-8}$). The signal lives in dimension-0 homology—the merging of well-served regions—rather than in enclosed “access deserts,” a finding that only emerges on real street networks.

1 Introduction

Target 11.7 of Sustainable Development Goal 11 commits states to provide “universal access to safe, inclusive and accessible, green and public spaces” [11]. Planners typically measure progress with catchment or proximity analyses: the share of residents within a walking threshold of a park. Such measures are static. They describe access on an ordinary day and are silent about *resilience*—how access to nature holds up when the pedestrian network is degraded by a flood, an earthquake, or the closure of a critical bridge or street. Two gaps follow. First, a city can score well on average proximity yet collapse under modest disruption; average accessibility does not reveal this fragility. Second, accessibility work rarely connects the *shape* of access to urban *form*, which is precisely the architecture–mathematics question that motivates this collection.

Persistent homology (PH) offers a principled, multi-scale, coordinate-free language for the “holes” and connected pieces of a coverage pattern [4, 8]. PH has been used to detect coverage gaps for amenities such as polling sites [6] and healthcare [5], to read the topology of urban congestion [12], and to quantify displacement under urban expansion [9]. What is *not* novel is using PH to find accessibility “holes” for an amenity. Our contribution is to move from *detecting* coverage gaps to *measuring the resilience of the access topology* and to *linking that resilience to urban form*.

Contributions.

- **A topological resilience metric** (the mathematics). We define $D(\rho)$, the Wasserstein distance between the persistent-homology diagram of the green-space accessibility field before and after disruption of intensity ρ , and summarise it by an area-under-curve (AUC) and a critical transition ρ^* (Section 4).
- **First topological treatment of SDG 11.7 green-space access** in an architecture–mathematics framing.
- **A district-level morphology↔resilience result** across 2,603 districts in five cities, with city fixed effects, showing that street-network morphology significantly predicts access resilience (Section 6).

2 Related work

TDA of spatial systems. Feng et al. [4] survey persistent homology for spatial data; Otter et al. [8] give the computational foundations. Within urban analysis, Wu et al. [12] read traffic congestion as a barcode, and a line of “resource coverage” work applies sublevel or network-distance filtrations to amenity access: polling places [6] and sexual-health services [5]. Rodríguez Vázquez and Cuerno [9] use PH to quantify displacement under urban expansion. These studies establish that PH can *locate* coverage holes. We differ on the object of study: we do not report where the deserts are, but how the whole access topology *degrades* under disruption, and we regress that degradation on morphology.

Accessibility and equity. The proximity/catchment tradition measures green-space access as a static field. Our accessibility field (Section 4) is built the same way—network walking time to the nearest green space—but is then read topologically and stress-tested, rather than thresholded once.

Street-network morphology. We characterise form with standard descriptors computed from OpenStreetMap via OSMnx [1]: intersection density, circuitry, orientation entropy, and mean street-segment length. Relating these to a resilience outcome is the architecture–mathematics bridge of this paper.

3 Mathematical background

Let $G = (V, E)$ be the pedestrian network and $f : V \rightarrow \mathbb{R}$ a scalar field giving, at each node, the network walking time to the nearest green-space access point. The *sublevel-set filtration* of f builds a growing simplicial complex $K_t = \{ \sigma : \max_{v \in \sigma} f(v) \leq t \}$. As the threshold t (walk-minutes) rises, well-served regions appear and merge. Persistent homology records, for each topological feature, a birth b and death d ; the multiset of points (b, d) is the *persistence diagram* $\text{Dgm}_k(f)$ in homological dimension k [3]. Dimension 0 tracks connected components (the merging of well-served regions); dimension 1 tracks loops (enclosed voids, i.e. access “deserts”).

The distance between two diagrams is measured by an optimal matching. The 1-Wasserstein distance is

$$W_1(X, Y) = \min_{\gamma} \sum_{x \in X} \|x - \gamma(x)\|,$$

minimised over bijections γ between the diagrams (with matching to the diagonal allowed). Wasserstein and bottleneck distances are stable to perturbations of f [2], which is what makes them a sound basis for a resilience metric. We compute PH with GUDHI [10] and diagram distances with **persim**.

4 Methodology

Data. For each city we take the pedestrian network and green/public spaces from OpenStreetMap via OSMnx [1], and we fix the analysis extent to the city’s *urban centre* polygon from the GHS Urban Centre Database [7] so that extents are comparable across countries. Green spaces are the OSM classes `leisure=park|garden|recreation_ground`, `landuse=grass|recreation_ground`, `place=square` and `natural=wood`; their centroids, snapped to the nearest network node, are the access points.

Accessibility field. We assign each node its network walking time (at 4.8 km/h) to the nearest green-space access point by multi-source Dijkstra, giving the field f .

Resilience metric. Let $\text{Dgm}(f)$ be the baseline diagram. A disruption of intensity $\rho \in [0, 1]$ removes a fraction ρ of network edges; we recompute the field and diagram on the disrupted network and define the degradation

$$D(\rho) = \mathbb{E}_r W_1(\text{Dgm}(f), \text{Dgm}(f_{\rho,r})),$$

averaged over r replicate disruptions. We consider three disruption families: uniformly random edge removal (percolation-style; used here), targeted removal of high-betweenness edges, and hazard removal of edges in low-elevation zones. The resilience of a place is summarised by the area under the curve $\text{AUC} = \int_0^{\rho_{\max}} D(\rho) d\rho$ (normalised by the ρ -range) and the critical transition ρ^* , the smallest ρ at which $D(\rho)$ first reaches half its maximum—a percolation-style threshold at which access structure breaks down.

Homology dimension and denoising. On real street networks the sublevel filtration’s finite H_1 is nearly empty: a street graph contains few triangles, so the network’s cycles are never filled and almost all H_1 classes are essential (infinite) and carry no finite persistence. In Amsterdam, for example, the baseline diagram has 6,449 finite H_0 classes but only a single finite H_1 class. The resilience signal therefore lives in *dimension 0*: the merging structure of well-served regions. We consequently compute $D(\rho)$ on H_0 . To keep the exact Wasserstein matching tractable on city-scale diagrams we apply persistence thresholding, discarding H_0 classes with lifetime below 1 walk-minute (topological noise); this preserves the ranking and ρ^* while reducing reported magnitudes by roughly 10–15%.

Districts and inference. Statistical inference rests not on five cities but on intra-urban *districts*: we tile each urban-centre boundary with a uniform H3 hexagonal grid (resolution 8, ≈ 0.7 km² cells) and compute, per district, its local resilience summaries and its local morphology (intersection density, circuitry, orientation entropy, mean segment length, and a green-space fragmentation index). We then fit

$$\text{AUC}_i = \beta^\top \mathbf{m}_i + \alpha_{c(i)} + \varepsilon_i,$$

an ordinary-least-squares regression of district AUC on morphology $\mathbf{m}_i$ with *city fixed effects* $\alpha_{c(i)}$ absorbing all between-city confounds. The five cities serve as a typological overlay, not as the inferential sample.

5 Case studies

We study five cities chosen to span street morphology, planning regime and Global North/South divide: İstanbul (organic, hilly, polycentric, seismic), Barcelona (Cerdà grid with superblock interventions), Amsterdam (compact, canal, walkable), Bogotá (dense informal-formal mix), and Phoenix

Table 1: District coverage (H3 resolution 8). Inference rests on $n = 2,603$.

City	Amsterdam	Barcelona	İstanbul	Bogotá	Phoenix	Total
Districts	227	147	729	538	962	2,603

Table 2: City typology (descriptive). Higher mean AUC $\Rightarrow$ access topology changes more under disruption $\Rightarrow$ *less* resilient.

City	mean AUC	mean ρ^*	mean H_0 total persistence
Amsterdam	4.25	0.27	13.11
Barcelona	6.54	0.29	13.63
Phoenix	6.58	0.18	8.08
İstanbul	7.44	0.21	12.84
Bogotá	7.54	0.24	16.90

(low-density grid sprawl). Boundaries are the GHS-UCDB urban-centre polygons; homonymous centres (e.g. Barcelona, Spain vs. Venezuela) are disambiguated by largest true (equal-area) extent.

6 Results

The random-disruption analysis yields 2,603 qualifying districts (Table 1); only 2.6% have zero AUC. District mean AUC ranks Amsterdam as the most resilient and Bogotá/İstanbul as the least (Table 2); Phoenix has the earliest breakdown ($\rho^* = 0.18$).

The headline result is the district-level regression (Table 3). With city fixed effects and $n = 2,603$, four of five morphology descriptors are significant, three at $p < 10^{-8}$. Because the coefficient sign is on AUC (higher AUC = less resilient), a negative coefficient means the descriptor is associated with *greater* resilience. More circuitous networks (coef -8.99), longer street segments (-0.048) and denser intersections (-0.199) are associated with more resilient green-space access, whereas higher orientation entropy ($+1.17$) is associated with less. Green-space fragmentation is not significant once it varies within cities and city fixed effects are included.

At the city scale, the Amsterdam degradation curve is monotone and smooth (Fig. 2): $D(\rho) = (0, 279.5, 1276.1, 1766.1)$ at $\rho = (0, 0.1, 0.3, 0.5)$, with AUC = 947.5 and $\rho^* = 0.30$, confirming that the metric responds meaningfully to disruption on real data.

7 Discussion

The results give an interpretable, architecture-facing reading of resilience. Locally redundant networks (those with denser intersections and, perhaps surprisingly, more circuitous routing) retain more of their well-served structure when edges are removed, because alternative short paths to green space survive. Highly ordered, low orientation-entropy districts are more brittle: a regular grid offers fewer topologically distinct detours once key links fail. That the signal is carried by H_0 rather than H_1 has a concrete planning meaning. In green-dense cities the structure that matters is not isolated “deserts” surrounded by served land but the connectivity of the served regions themselves, and that connectivity is what disruption erodes.

Table 3: District fixed-effects regression, $\text{AUC} \sim \text{morphology} + C(\text{city})$, $n = 2,603$, random-disruption scenario. Sig.: *** $p < 0.001$, ** $p < 0.01$.

Morphology descriptor	coef	std. err.	p -value
circuitry	-8.991	1.185	4.5×10^{-14} ***
orientation entropy	+1.172	0.203	9.4×10^{-9} ***
mean street length	-0.0478	0.0059	7.2×10^{-16} ***
intersection density	-0.199	0.076	8.6×10^{-3} **
green-space fragmentation	-1.5×10^{-6}	3.8×10^{-6}	0.70

Limitations. (i) We report the random-disruption scenario; targeted (betweenness) and hazard (low-elevation) scenarios are implemented but not yet run at scale, and coefficient signs should be checked for stability across them. (ii) Results use a persistence-thresholded H_0 metric and a reduced disruption grid (5 intensities, 3 replicates); magnitudes are therefore denoised estimates, though ranking and ρ^* are stable. (iii) City-level degradation curves (Fig. 5) are so far computed for Amsterdam only. (iv) OSM completeness varies by city, and the network-distance Vietoris–Rips variant of the filtration is left to future work. None of these undermines the core finding: at the scale of thousands of districts, street-network morphology significantly predicts the resilience of green-space access.

8 Conclusion

We defined a topological resilience metric for green-space accessibility—the Wasserstein degradation curve of persistent-homology diagrams under network disruption—and showed, across 2,603 districts in five contrasting cities, that urban morphology significantly predicts how robustly a city keeps its population close to nature. The mathematics of persistence turns a static SDG 11.7 indicator into a dynamic, form-aware one. Future work will complete the targeted and hazard scenarios, add a network-metric filtration, and test resolution sensitivity, toward a resilience-aware accessibility standard for sustainable cities.

Reproducibility. All data are open (OpenStreetMap, GHS-UCDB) and the pipeline is open-source Python (OSMnx, NetworkX, GeoPandas, GUDHI/persim, H3, statsmodels); every reported number is recorded in the project’s results log and traces to a committed artifact.

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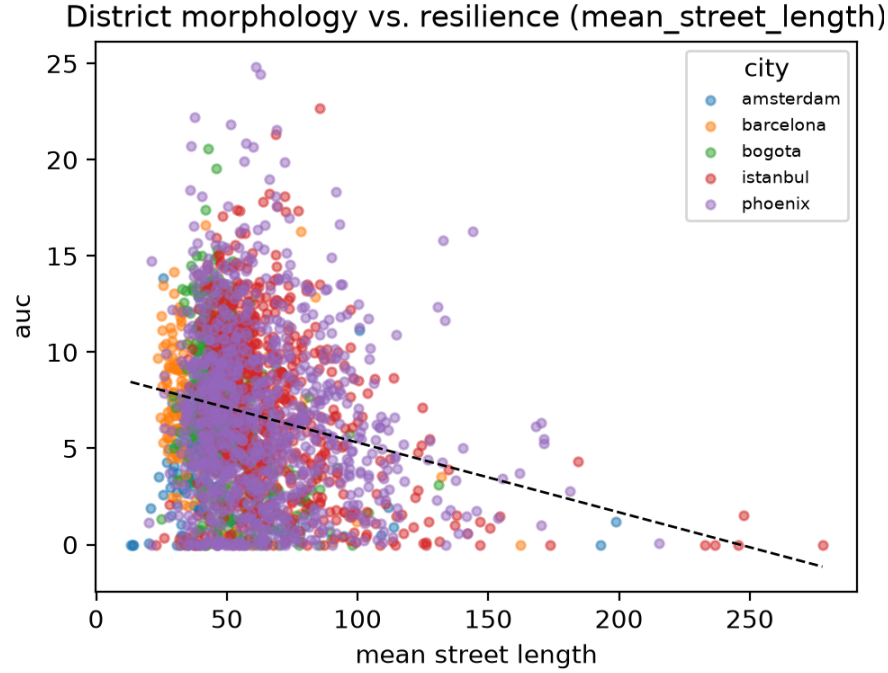


Figure 1: Headline morphology $\leftrightarrow$ resilience relationship. Each point is one of the 2,603 districts, coloured by city; the dashed line is the pooled least-squares fit. Higher AUC indicates a district whose green-space access topology degrades more under disruption (less resilient).

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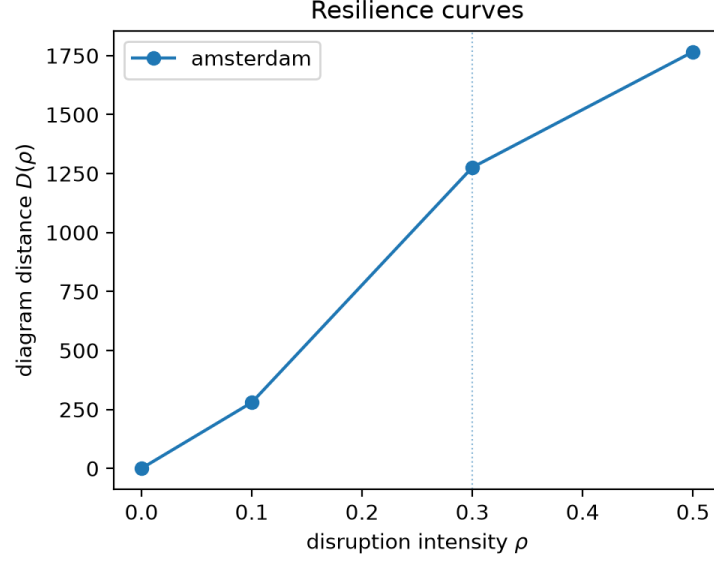


Figure 2: City-level resilience curve $D(\rho)$ for Amsterdam (random-disruption scenario). The dotted vertical line marks the critical transition ρ^* .

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